BPUT HACKATHON 2026

Sustainable Facility and Estate Intelligence Platform for India

From Data to Decisions for Smarter, Sustainable Campuses



PS CODE : PS-04

TEAM NAME : TEAM ELITE

TEAM ID : BH26PS04T039



From Facility Data to Intelligent Decisions

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One platform that answers the questions facility teams ask every day — before the bill arrives.

01
THE PROBLEM
What is going wrong?

02
WHO CARES
Who feels the pain?

03
THE SOLUTION
What are we building?

04
HOW IT WORKS
Architecture & demo

05
WHY US
Research gaps — our edge

06
BUSINESS
Can it make money?

07
IMPACT
Why it matters

08
WHAT'S NEXT
Roadmap

The Problem

Facilities Generate Data. Decisions Still Happen Manually.

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Situation

Current Reality

Water Data
Separate meters, different formats

Energy Data
Multiple meters, buildings, transformers

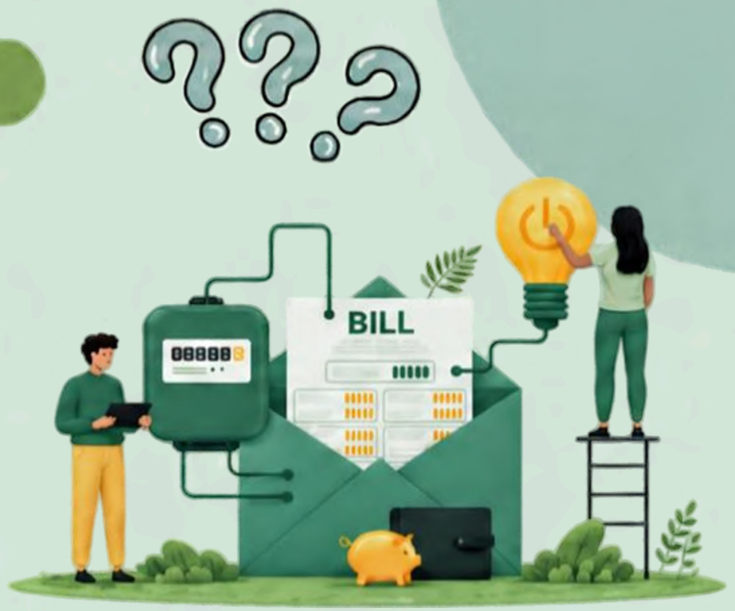
Maintenance
Mostly manual, reactive

Analysis
Done manually in spreadsheets

Decision
Human dependent and delayed

Problem





The problem is not only resource consumption. The problem is fragmented visibility and delayed decision-making.



WHO FACES THE PROBLEM?

Built for the People Who Run Facilities

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COLLEGES
GRIHA mandates, tight budgets



HOSPITALS
24/7 energy, patient safety



CORPORATE CAMPUSES
ESG reporting pressure



PSUS & GOVT OFFICES
Mandatory energy audits



INDUSTRIAL ESTATES
Resource-heavy operations



MUNICIPAL FACILITIES
Smart city programs

THEIR DAY SOUNDS LIKE THIS

FACILITY ADMINISTRATOR “TOO MUCH DATA, TOO LITTLE VISIBILITY.”

OPERATIONS TEAM “WHICH ISSUE NEEDS ATTENTION FIRST?”

SUSTAINABILITY OFFICER “ARE WE ACTUALLY IMPROVING?”

MANAGEMENT “WHAT HAPPENS IF WE CHANGE THIS?”



RESEARCH PAPER STUDY

RESEARCH GAPS → OUR ENGINEERING ADVANTAGES

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P1 — HIMEUR ET AL. 2021 (APPLIED ENERGY)

OBJECTIVE

TAXONOMY OF AI ANOMALY-DETECTION METHODS; TRENDS & PERSPECTIVES

METHODOLOGY

TAXONOMY OVER ML MODEL, FEATURES, DETECTION LEVEL, PLATFORM, APPLICATION, PRIVACY

LIMITATIONS

NO UNIFIED METRICS; NO REPRODUCIBILITY PLATFORM; ANOMALIES ILL-DEFINED; LABELED DATA ABSENT; PRIVACY RARELY ADDRESSED

RESEARCH GAPS

(I) PRECISE ANOMALY DEFINITION
(II) ANNOTATED DATASETS
(III) UNIFIED METRICS
(IV) REPRODUCIBILITY PLATFORMS
(V) PRIVACY PRESERVATION

P2 — BHARATH ET AL. 2026 (DISCOVER SUSTAINABILITY)

OBJECTIVE

EXPLAINABLE MULTI-ZONE DIGITAL TWIN WITH QUANTIFIED DECISION IMPACT

METHODOLOGY

XGBOOST REGRESSION + SHAP TREEEXPLAINER + SCENARIO SIMULATOR + PAIRED WITH/WITHOUT-XAI TRIALS (N=42)

LIMITATIONS

ALGORITHM-ONLY TRIALS (NO HUMANS); SINGLE BUILDING/CLIMATE; CVRMSE 45.7% > ASHRAE 30%; XAI POLICY 24% FAILURE; ROLE UI UNTESTED

RESEARCH GAPS

HUMAN DECISION-EFFECTIVENESS UNTESTED; CROSS-ZONE COUPLING LIMITED; XAI CONFIDENCE-THRESHOLDING NEEDED

P3 — MUNIR ET AL. 2026 (IET SMART GRID)

OBJECTIVE

SURVEY OF AI/DL/ML FORECASTING METHODS FOR BUILDING ENERGY

METHODOLOGY

COMPARATIVE ANALYSIS OF PHYSICAL, DATA-DRIVEN, HYBRID FORECASTING

LIMITATIONS

REVIEW-LEVEL ONLY; INCONSISTENT DATA FORMATS/HORIZONS; TECHNOLOGY-ORIENTED RATHER THAN PROBLEM-DRIVEN RESEARCH

RESEARCH GAPS

UNSTRUCTURED/NON-STANDARD DATA; 128+ ALGORITHMS, NO DOMINANT ONE; BENCHMARK STANDARDS NEEDED

PROPOSED SOLUTION

MORE THAN A DASHBOARD

EstateIQ

01

AI Anomaly Detection

Spots unusual energy, water behaviour — after checking occupancy, weather and time of day.

02

Short-Term Forecasting

Predicts the next 24 hours to 7 days of demand, so teams act before problems arrive.

03

Explainable AI

Every alert says WHY. SHAP shows which factor drove it — no black boxes.

04

What-If Simulation

“What if we cut HVAC 10%?” Test decisions safely, see estimated impact first.

ELITE ALGORITHM

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01 | TELEMETRY INPUTS

- Building Meters
- Occupancy (PPL)
- Ambient Temp (°C)
- HVAC Schedules

02 | QUALITY & SCREENING

STAGE 1: DATA QUALITY GATE

- Completeness
- Freshness
- Sensor Reliability

STAGE 2: CONTEXT FILTER

EXECUTION PATHS

- Path A: Fast
- Path B: Intelligence
- Path C: Critical

03 | FORECASTING & ANOMALIES

STAGE 3: ADAPTIVE ENSEMBLE

Champion-Challenger

Elite Algorithm

Adaptive forecasting

STAGE 4: ANOMALY CONSENSUS

Multi-Signal &

Anomaly Score

04 | EVIDENCE TRUST & IMPACT

STAGE 5: CONFIDENCE ENGIIE

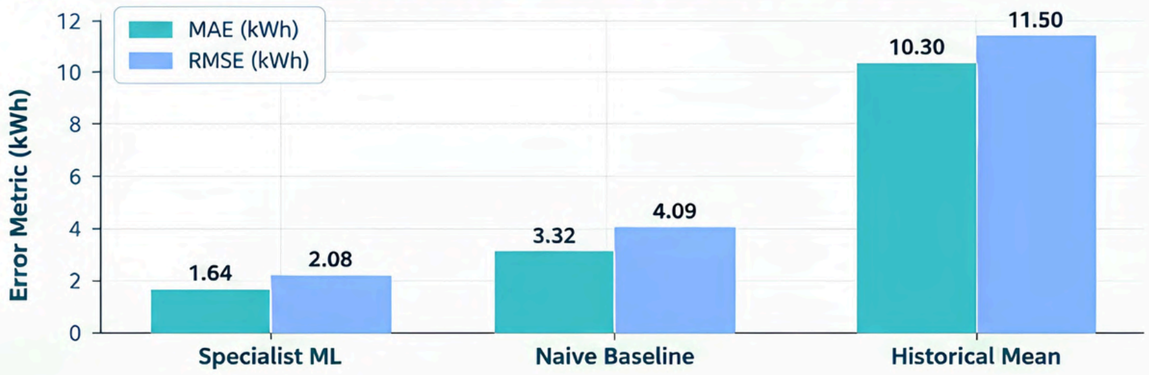
Trustworthiiess Score

- Data Quality
- Model Agreiment
- History
- Compateness
- Reliabiliy

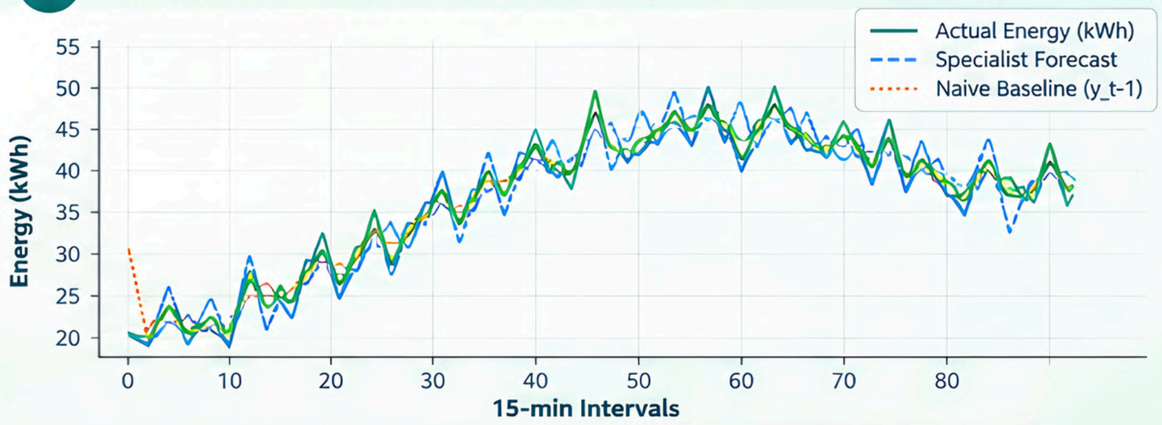
ALGORITHM METRICS

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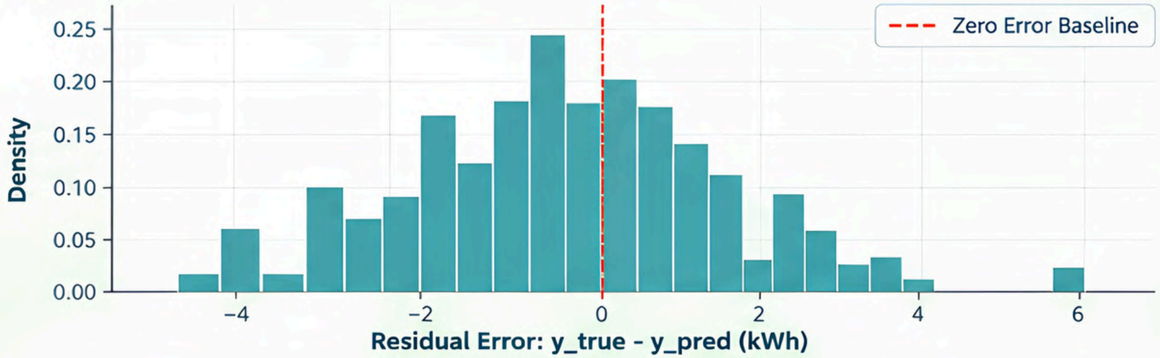
A Model Performance Comparison (MAE vs RMSE)



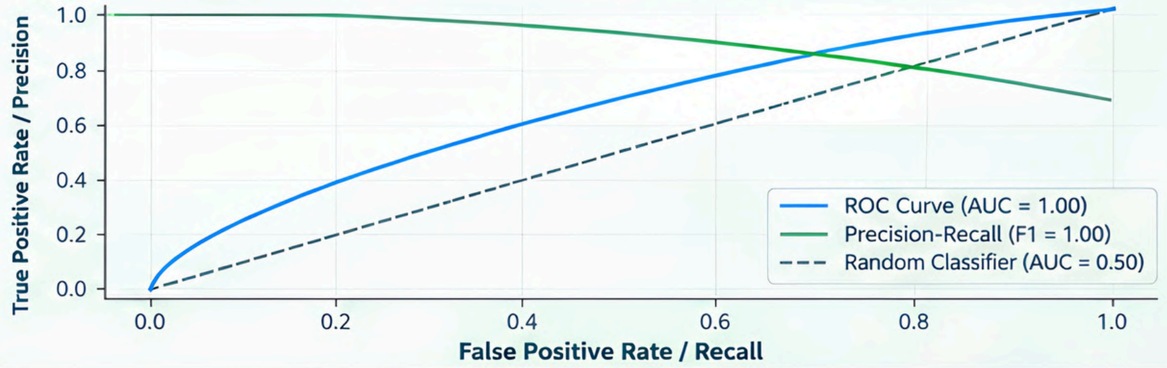
B Actual vs Predicted Energy Consumption



C Residual Error Distribution (Mean: -0.40, Std: 2.04)



D ROC & Precision-Recall Anomaly Classifier Curves



ELITE ALGORITHM — COMPUTATIONAL COMPLEXITY

Decision-layer Time Complexity

$$O(M + K \log K + F)$$

Combines processing of specialist model signals, action ranking and contextual features.

Auxiliary Space Complexity

$$O(M + K + F)$$

Stores model outputs, candidate actions and contextual features.

Where:

- M = number of specialist-model signals (e.g., forecast, anomaly scores)
- K = candidate actions / recommendations being ranked
- F = contextual features used in fusion/scoring (e.g., asset type, operational context, cost factors)

Elite Algorithm Decision Pipeline

Validate Data → Fuse Signals → Score Impact → Rank Decisions → Recommend Actions

⚠ Illustrative upper bound under these assumptions; excludes model training/inference and depends on implementation.

Evaluation values must be validated on the current test dataset.

Tech Stack

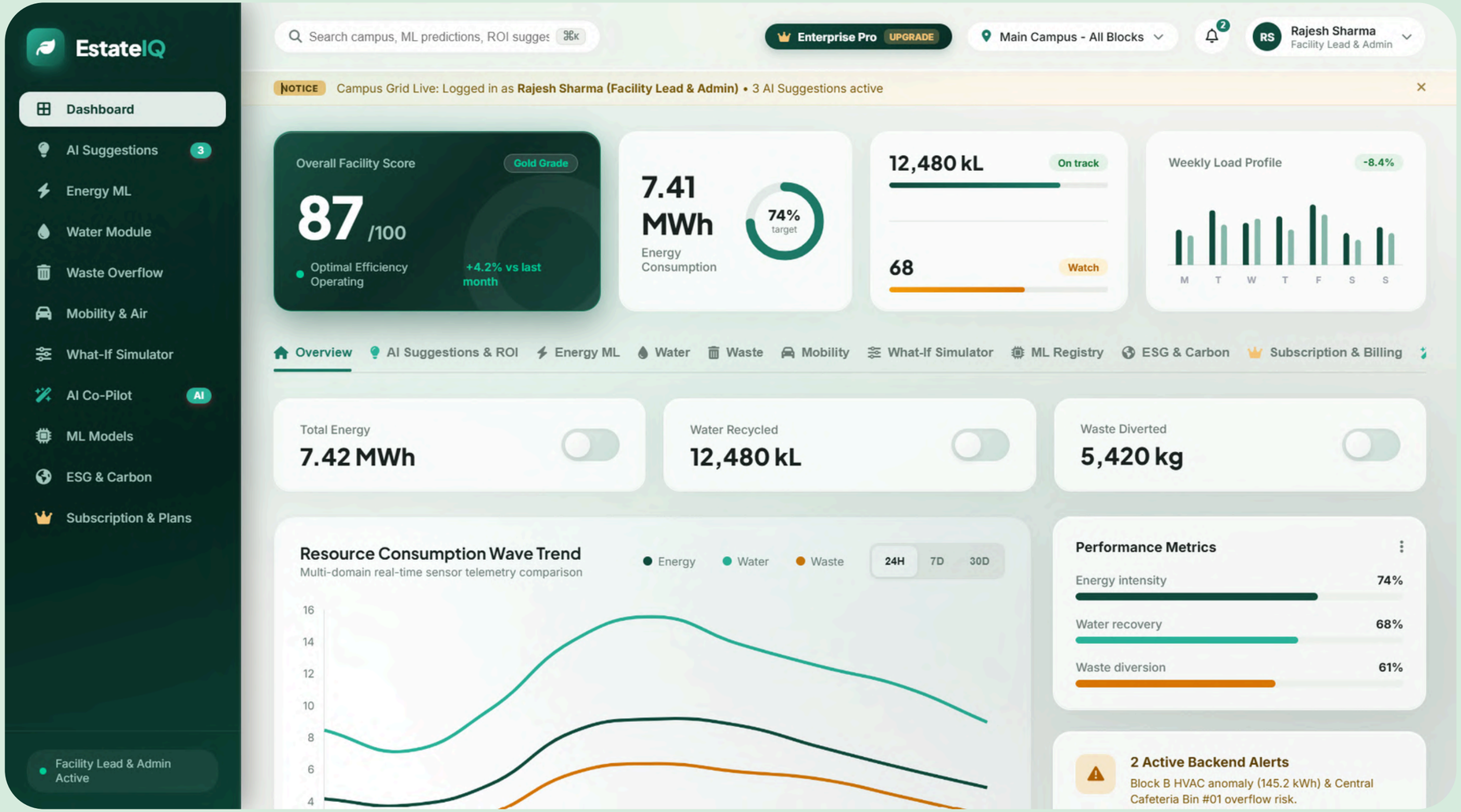
Built to Be Practical, Not Just Impressive

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Category	Technologies
Core & Backend	Python 3.10+, FastAPI, Uvicorn, Pydantic
AI / ML	Scikit-learn, XGBoost, LightGBM, CatBoost, Prophet
Intelligence	SHAP, Anomaly Detection, Forecasting, Optimization
Data & Visualization	Pandas, NumPy, Plotly, GeoPandas, Folium
Frontend	Streamlit
IoT	**Electricity:** Raspberry Pi, Three Phase Meter, Circuit Transformer **Water Sensor:** Raspberry Pi, Water Flow Meter, Tank Level Transmitter
Security	Advanced Encryption Standard, JWT, Role-Based Authentication

PRODUCT DEMO

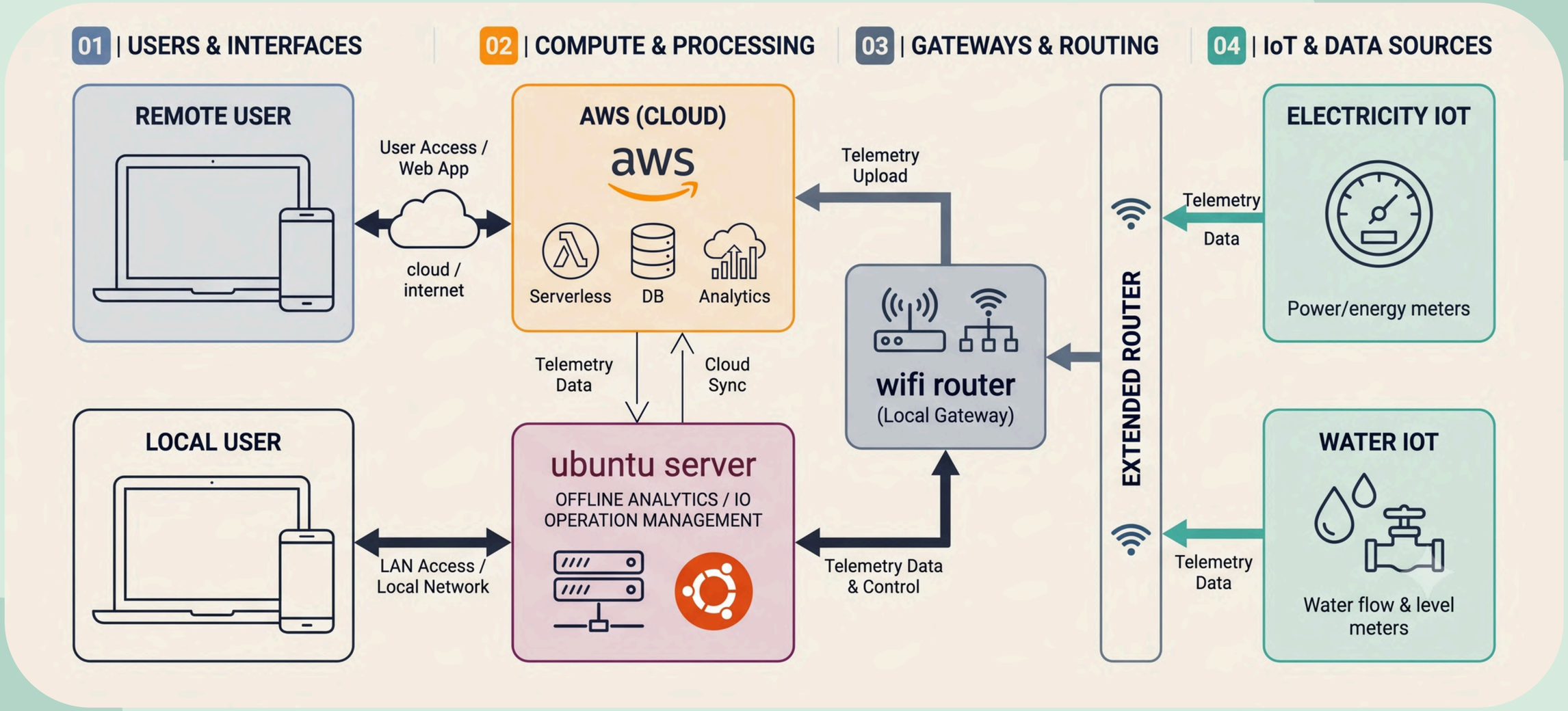
From Alert to Action



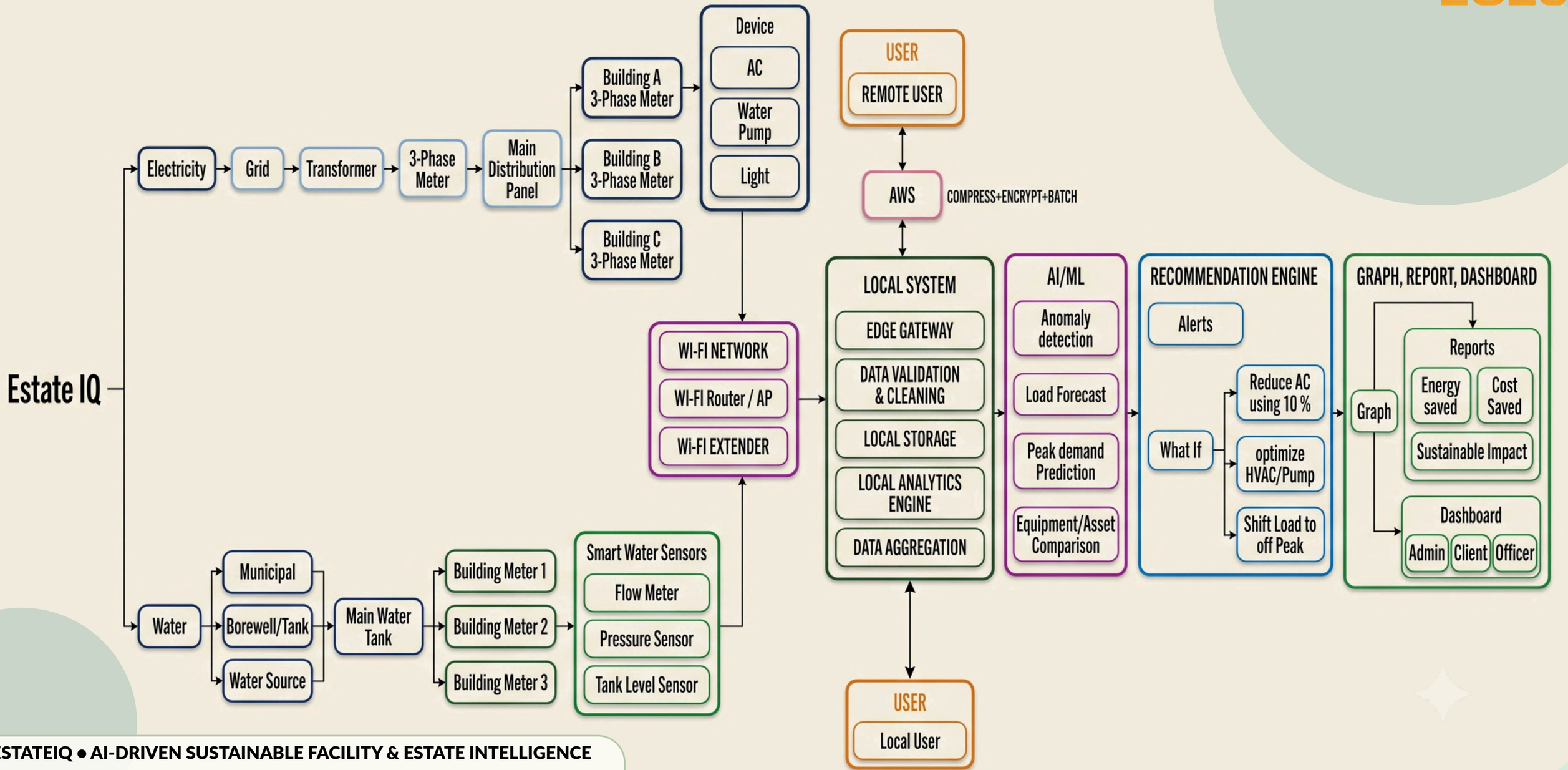
WHAT THE ADMIN JUST DID

1. **SELECTS FACILITY**
   ONE DROPDOWN — COLLEGE, HOSPITAL, PSU...
2. **SEES THE ALERT**
   SEVERITY-RANKED, NOT A WALL OF CHARTS
3. **READS THE WHY**
   SHAP NAMES THE DRIVER: HVAC 47%
4. **CHECKS THE FUTURE**
   24H FORECAST WITH CONFIDENCE BAND
5. **TESTS A FIX**
   WHAT-IF: −9% ENERGY, COMFORT INTACT
6. **DECIDES**
   ACCEPT → ACTION LOGGED IN AUDIT TRAIL

ARCHITECTURE DIAGRAM



Flow Chart


Estate IQ
Electricity
Grid
Transformer
3-Phase Meter
Main Distribution Panel
Building A 3-Phase Meter
Building B 3-Phase Meter
Building C 3-Phase Meter
Device
AC
Water Pump
Light
Water
Municipal
Borewell/Tank
Water Source
Main Water Tank
Building Meter 1
Building Meter 2
Building Meter 3
Smart Water Sensors
Flow Meter
Pressure Sensor
Tank Level Sensor
WI-FI NETWORK
WI-FI Router / AP
WI-FI EXTENDER
USER
REMOTE USER
AWS
COMPRESS+ENCRYPT+BATCH
LOCAL SYSTEM
EDGE GATEWAY
DATA VALIDATION & CLEANING
LOCAL STORAGE
LOCAL ANALYTICS ENGINE
DATA AGGREGATION
USER
Local User
AI/ML
Anomaly detection
Load Forecast
Peak demand Prediction
Equipment/Asset Comparison
RECOMMENDATION ENGINE
Alerts
What If
Reduce AC using 10 %
optimize HVAC/Pump
Shift Load to off Peak
GRAPH, REPORT, DASHBOARD
Graph
Reports
Energy saved
Cost Saved
Sustainable Impact
Dashboard
Admin
Client
Officer

BUSINESS MODEL

Hardware once, platform your way, AI as a paid upgrade

For colleges · hospitals · corporate campuses · PSUs · industrial estates · municipal facilities

1 HARDWARE

- OEM partners
- Meters, sensors, install

One-time + service

2 PLATFORM · THEIR CHOICE

- On-site: local server, offline
- Cloud: AWS, access anywhere

Licence or monthly plan

3 AI ADD-ON

- Forecasts, SHAP "why"
- What-if, ranked fixes

Paid, recurring

Our Advantages	Manual audits	Traditional BMS	Generic energy SaaS	EstateIQ
Energy, water in one	No	Partial	Partial	**Yes**
Explains why (SHAP)	No	No	Limited	**Yes**
What-if before acting	No	No	Limited	**Yes**
Verifies result after action	No	No	Limited	**Yes**
Works offline	n/a	Yes	Cloud only	**Yes + cloud option**
Upfront cost	Labour-heavy	High	Medium	**Low**

COST-EFFECTIVE BY DESIGN

- Raspberry Pi-based edge hardware
- Start with one building
- Pay for AI only when needed

Others stop at data. EstateIQ ranks, explains, tests and verifies, and every fix shows its ₹ impact (ΔE × T × ₹9.50/kWh).

FUTURE SCOPE

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From decision support to a trusted, scalable estate platform

TODAY

EstateIQ monitors, predicts, explains and recommends. The admin decides.

TOMORROW

EstateIQ pinpoints faulty, energy-wasting equipment, recommends efficient replacements, and verifies savings after upgrades—with admin approval

SHORT TERM — Validate

- Pilot in a real campus
- Human-in-the-loop trials
- Benchmark accuracy against industry standards

MID TERM — Expand

- Multi-building, cross-zone coupling
- Air quality, waste and mobility modules
- Partner hardware rollout, automated ESG reports

LONG TERM — Scale

- Portfolio view across campuses and cities
- Privacy-preserving learning across sites

MORE RESOURCES

Air quality | Waste management | Mobility and fleets | Carbon and renewables

MORE SECTORS

Healthcare | Education | Industry | Smart cities | Government

CONCLUSION

Facilities don't lack data.
They lack decisions.

THE PROBLEM

Fragmented data

- Separate meters
- Manual analysis
- Late decisions

OUR ANSWER

Decisions, explained

- Predicts and explains
- Tests the fix first
- Verifies the result

THE EDGE

Affordable, flexible

- On-site or cloud
- Pay for AI when needed
- Grows to air and waste

Monitor. Explain. Decide. Before the bill arrives.

Thank you · Team Elite · Questions welcome